Towards Fair Fares: Designing a Resilient Contactless Payment Ecosystem for Public Transit in Nepal

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Urban mobility in rapidly growing Nepali cities like Kathmandu and Pokhara is severely constrained by a persistent reliance on cash-based fare collection. Recent attempts to digitize public transit using general-purpose QR code wallets (such as *eSewa* and *Khalti*) have inadvertently introduced new infrastructural frictions. These mobile-first solutions often fail in the dynamic setting of a moving vehicle, where scanning delays and dropped cellular signals exacerbate boarding bottlenecks and deepen existing digital divides. To address this socio-technical gap, we introduce *TransitPay*, an equitable, offline-capable payment ecosystem using Near Field Communication (NFC) and Host Card Emulation (HCE). By intentionally designing around affordable hardware (ESP8266 and PN532) and utilizing a local data queuing architecture, the system processes transactions independently of real-time cloud connectivity. Importantly, shifting the internet requirement exclusively to the vehicle's Point-of-Sale terminal removes the burden of mobile data access from the commuter and aligns fare collection with the realities of resource-constrained transit infrastructure. Sandboxed evaluations demonstrate that *TransitPay* reduces transaction latency from a typical 30 seconds to just 1.5 to 2.5 seconds. This 94 percent reduction in wait time represents more than a technical optimization; it marks a step toward inclusive, resilient digital public infrastructure for everyday urban mobility in the Global South.

CCS Concepts: • **Applied computing** → **Transportation**; • **Computer systems organization** → **Embedded and cyber-physical systems**; • **Human-centered computing** → *Ubiquitous and mobile computing systems and tools*.

Additional Key Words and Phrases: Urban Mobility, Digital Equity, Contactless Transit, Offline-First Architecture, Host Card Emulation

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1 Introduction

Consider a commuter navigating a crowded microbus in Kathmandu. In a valley where over 90% of the public transit fleet consists of low-capacity minibuses and microbuses, the conductor must squeeze through a packed aisle to collect physical currency [4, 19]. This manual process routinely leads to fare disputes and severe revenue leakage, with reports indicating that conductors often hand over only half of their actual daily collections to vehicle owners [36]. Because Nepal's public transit remains almost entirely dependent on cash, the entire urban mobility network suffers from severe boarding bottlenecks. When the payment

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process stalls, the daily dignity and economic mobility of the 28% of valley residents who rely on this infrastructure are significantly restricted [19].

While digital wallets like *eSewa* and *Khalti* have successfully modernized retail in Nepal, public transit presents a uniquely hostile environment for standard mobile payments. The core issue is severe boarding latency. Commuters attempting to scan a QR code face transaction times that can drag on for up to 30 seconds, often while standing in crowded, moving vehicles. More importantly, these mobile-first systems require continuous cellular data and completely fall apart when a vehicle enters a network blind spot [15, 21, 22]. In a context where broadband connectivity is highly inconsistent, these digital attempts frequently fail in practice, forcing commuters and conductors back to manual cash handling and deepening existing digital divides.

Previous attempts to digitize transit fares in the region have recognized the need for faster technologies but have seen limited success. Initiatives like Sajha Yatayat's smart cards [36] and the Yatra Card [10] introduced NFC technology for a faster tap-in and tap-out experience. However, these implementations operate strictly as closed-loop systems. A proprietary card for one bus network cannot be used on a private microbus, failing to solve the interoperability gap across fragmented, semi-formal fleets. Furthermore, while the National Payment Switch integrates retail wallets, it currently lacks a standardized API for cross-operator transit fare settlement, limiting the emergence of transit payments as shared digital public infrastructure.

To bridge this socio-technical gap, we detail the development of TransitPay, an equitable, offline-capable payment ecosystem built using Near Field Communication (NFC) and Host Card Emulation (HCE). Rather than assuming always-on cloud connectivity on personal devices, TransitPay relocates computational and connectivity burdens to affordable, vehicle-mounted edge infrastructure. By intentionally designing around simple, low-cost hardware, i.e., the ESP8266 [9] and PN532 [28], and utilizing a local data queuing architecture, the system processes transactions securely offline and synchronizes with the cloud only when the vehicle regains cellular range. To enable economic fairness, the system also integrates a GPS module to track boarding and alighting coordinates automatically, allowing fares to be computed from a transparent, distance-based matrix.

This work is guided by two primary research questions:

- **RQ1**: How can an offline-first NFC architecture reduce boarding latency and improve payment resilience in resource-constrained transit environments?
- **RQ2**: How does an automated, location-based payment model impact the transparency and economic fairness of fare collection for daily commuters?

We position TransitPay as a case of urban digital public infrastructure that is explicitly designed for sustainable and equitable societies, rather than as a mere optimization of retail payment rails. Drawing on these research questions, we offer three main contributions to the field of sustainable urban computing and computing for sustainable societies:

- The design and implementation of TransitPay, an affordable (NPR 3000) IoT hardware architecture that demonstrates how offline-first, edge-based designs can bypass intermittent internet connectivity via local data queuing in informal transit systems.

- The introduction of a "Fair Fare Model" that utilizes onboard GPS coordinates to automatically calculate distance-based pricing, eliminating manual revenue leakage and protecting passengers from arbitrary overcharging, thereby operationalizing fare transparency as a form of digital equity.
- Sandboxed empirical evaluations showing that this system achieves a best-case transaction speed of 1.5 to 2.5 seconds, reducing wait times by 94% compared to existing QR methods, and offering design implications for latency-sensitive civic payment infrastructures in similarly resource-constrained contexts.

2 Related Works

2.1 Digital Fare Collection in the Global South

Across the Global South, public transportation frequently relies on decentralized, semi-formal fleets like minibuses, jeepneys, or matatus [19]. The transition from cash-based fare collection to digital payment ecosystems in these environments presents substantial infrastructural and governance challenges. While QR code payments have historically dominated the broader retail sectors in large markets such as India and China [8], deploying these optical systems in dynamic, moving transit environments often introduces bottlenecks and exacerbates existing access gaps [8].

Alongside these broader payment trends, recent Information and Communication Technology for Development (ICTD) work has examined how riders in the region navigate tensions between informal, cash-based services and app-based platforms. For example, Alok et al. document why many commuters in Bangladesh still prefer contractual "Khep" rides over ride-sharing apps, citing factors such as bargaining practices, trust, and affordability [3]. Complementary work on public transit advocates for socio-technical toolkits that foreground equity and rider experience: the BALTO Toolkit project develops ethical and sustainable data-collection methods to help communities document service gaps and advocate for fairer public transit [12].

Currently, there is a notable shift toward NFC technologies in developing regions to address these frictions. Southeast Asia is experiencing a massive surge in NFC adoption, with countries like the Philippines, Thailand, and Vietnam projected to reach 75 percent digital wallet penetration by late 2026 [20]. As these nations attempt to integrate digital payments into their transit networks, direct comparisons between QR and NFC methods have become vital [11]. Research on mobile fare payment systems shows that QR-code ticketing can increase boarding service time due to scanning procedures and recognition delays, sometimes causing crowding at transit entry points, whereas NFC-based "tap-and-go" systems validate transactions in around 100-120 ms and enable faster passenger throughput [7]. TransitPay extends this body of work by demonstrating how a low-cost, NFC-based architecture can be tailored to the constraints of semi-formal fleets in Nepal while still delivering substantial reductions in boarding latency.

2.2 Digital Public Infrastructure and the Political Economy of Informal Transit

The modernization of transit networks in the Global South is increasingly framed through the lens of Digital Public Infrastructure (DPI) [33]. Foundational systems such as India's Unified Payments Interface (UPI) and Kenya's M-Pesa have successfully democratized financial access in the retail sector by leveraging widespread mobile network penetration [14, 34]. However, applying these DPI frameworks directly to informal transit

reveals socio-technical frictions [33, 41]. Standard mobile wallets are optimized for static retail environments where a transaction delay of ten to twenty seconds is acceptable. In the dynamic environment of urban transit, these cloud-dependent architectures struggle to meet the high-throughput demands of boarding passengers, particularly in geographic areas with intermittent cellular coverage [41].

Furthermore, deploying digital payment systems in informal transit requires navigating a complex political economy. Semi-formal networks, such as the microbus syndicates in Kathmandu or the matatu fleets in Nairobi, operate on a highly localized, cash-based micro-economy [29, 35]. Conductors and drivers rely on immediate liquid cash to cover daily operational expenses, including fuel, route fees, and their own wages [26]. When digital payment systems introduce settlement latency of 24 to 48 hours, they inadvertently create severe cash flow constraints for hand-to-mouth enterprises.

Because of this liquidity delay, transit crews often actively resist the adoption of cashless systems. Purely technical interventions frequently fail when they ignore these entrenched financial realities. To be sustainable, transit technologies must be accompanied by socio-economic alignments, such as rapid settlement cycles and performance-based micro-incentives for conductors [26, 35]. TransitPay is positioned within this broader context. By acknowledging the limitations of cloud-dependent DPI in transit environments, the architecture prioritizes edge-based resilience while creating a technical foundation that can ultimately support the rapid batch settlements required by informal transit operators.

2.3 The Offline-First Imperative for Financial Inclusion

A core challenge in deploying transit fintech in the Global South is the assumption of continuous, ubiquitous internet connectivity [37]. Many modern mobile wallets rely entirely on real-time cloud authorization to process transactions [6, 37]. This design choice inherently introduces friction in resource-constrained environments, creating a barrier to true financial inclusion for users who cannot afford consistent mobile data plans or who are traveling through network dead zones. In Nepal, for instance, weak last-mile connectivity and patchy broadband heavily restrict the reliability of real-time cloud payments on moving vehicles [23].

To combat infrastructural fragility, researchers in ICTD increasingly advocate for "offline-first" architectures [18, 38]. By localizing data processing to the edge device, systems can ensure transaction resilience even when the broader network fails [18]. TransitPay integrates this philosophy directly into its hardware and software design. By utilizing a local data queuing method on the ESP8266 microcontroller, the system deliberately shifts the connectivity burden away from the commuter's smartphone and onto the vehicle's Point-of-Sale terminal. This localized processing ensures that digital public goods remain accessible, equitable, and functional, regardless of fluctuating cellular signals.

2.4 Digital Transit Infrastructure in Nepal

To address these global challenges locally, recent initiatives in Nepal have attempted to digitize urban transit using Near Field Communication (NFC). NFC is a short-range wireless technology that enables secure, instant data exchange between active and passive devices separated by just a few centimeters, making it the standard for frictionless "tap-and-go" payments [32]. In Kathmandu, operators like Sajha Yatayat and Fintech International have introduced NFC-enabled smart cards (such as the Yatra Card) to calculate distance-based fares and reduce revenue leakage [10, 36].

Beyond fare media, related work has begun to treat transit and mobility infrastructure as shared digital public goods. Bolten and Caspi, for instance, describe a communal, open-data approach to constructing detailed pedestrian networks to support accessibility analysis and inclusive routing at city scale [5]. Similarly, the BALTO Toolkit positions rider-generated data as a basis for more equitable transit planning and community advocacy in U.S. cities [12]. We build on this perspective by proposing fare-collection infrastructure that is designed from the outset to be interoperable across operators and to reduce information asymmetries between commuters, conductors, and vehicle owners.

However, existing NFC-based solutions in Nepal operate strictly as closed-loop systems. A proprietary card issued for one network cannot be utilized on a private microbus, failing to solve the interoperability gap for the average commuter, especially since the National Payment Switch currently lacks a standardized transit API [27]. To bypass this fragmentation, commuters frequently attempt to use ubiquitous mobile wallets (such as *eSewa* or *Khalti*) that rely on QR codes. Unfortunately, direct comparisons reveal that manual QR scanning in dynamic transit environments routinely takes over 10 seconds, causing severe passenger stacking at bus doors [7]. Furthermore, because these mobile-first systems lack offline data queuing, they frequently fail when moving vehicles cross network blind spots [23].

TransitPay builds upon these foundational efforts by utilizing Host Card Emulation (HCE) to emulate NFC cards via software [1, 2]. This approach merges the interoperability of a mobile application with the offline resilience and speed of a physical NFC smart card, engineering a solution specifically tailored for Nepal's fragmented and connectivity-constrained transit ecosystem.

3 Formative Study and Local Context

The core premise of TransitPay is that sustainable urban computing must directly address local human friction. To transition this project from a theoretical payment gateway into a practical socio-technical intervention, we first needed to map the lived experiences of Nepal's transit users. Through a human-centered design paradigm, we investigated the specific infrastructural limitations and behavioral anxieties that currently block the adoption of digital transit fares.

3.1 The Kathmandu Transit Ecosystem

To contextualize the commuters' experiences, it is essential to understand the physical and operational realities of Kathmandu's informal transit network. The city relies almost entirely on decentralized, semi-formal fleets consisting primarily of 14-seater microbuses, 26-seater minibuses, and three-wheeled electric Safa Tempos [19]. These vehicles are largely managed by independent operators organized into route-based syndicates [39].

The operational model relies heavily on a two-person crew: a driver and a conductor, locally referred to as a *khalasi* or *sahalachalak*. The conductor is responsible for recruiting passengers by calling out destinations, maximizing vehicle occupancy, and physically navigating an often severely overcrowded aisle to collect cash fares. During peak hours, passenger density routinely exceeds the vehicle's designed capacity. Commuters are frequently forced to stand tightly packed in narrow, low-ceilinged aisles. Within this highly compressed physical environment, the act of payment is not merely a financial transaction but a spatial challenge. Any payment method that introduces latency, whether it is a passenger fumbling for loose change or waiting for a cloud-dependent QR application to authenticate over a weak cellular signal, directly exacerbates the

physical discomfort of the entire cabin. This socio-spatial reality necessitates payment interventions that prioritize absolute speed and physical convenience.

3.2 Methodology

We conducted a preliminary, mixed-method formative study targeting regular and former public transportation users in Kathmandu. The study utilized a structured survey instrument (N=32) designed to capture both quantitative metrics and qualitative narratives regarding the current cash-based and QR-based payment ecosystems. The survey consisted of closed-ended Likert-scale questions to measure user acceptance of digital wallet integration, alongside open-ended questions allowing respondents to describe their personal experiences with cash handling, conductor interactions, and physical crowding.

The demographic breakdown consisted of 26 undergraduate students, one senior citizen, and five non-student commuters. Among the respondents, 24 utilize public transit regularly. The remaining eight participants reported reducing or completely abandoning their public transit usage, frequently citing systemic frictions as the primary deterrent.

3.3 Identified Frictions and Problem Classification

Through thematic analysis of the survey responses, we classified the commuters' challenges into three primary socio-technical frictions, using an inductive coding approach. The open-ended survey responses were systematically reviewed to identify recurring keywords and emotional indicators associated with the commute. Initial codes included recurring phrases such as "waiting for change," "network failure," "fake student ID," and "argument over fare."

These initial codes were then synthesized into broader thematic categories. Instead of treating these issues as isolated operational complaints, we mapped these qualitative themes to extract the root causes of transaction failures. These identified problems directly informed the engineering requirements for the TransitPay architecture, allowing us to translate subjective user frustrations into objective hardware and software interventions.

3.3.1 Friction 1: The Boarding Bottleneck and Cash Dependency. The most prominent challenge reported by users was the physical discomfort and severe latency caused by manual cash transactions inside crowded vehicles. Our quantitative data reveals that 22 out of the 24 regular commuters consistently faced cash-related problems with conductors. Commuters highlighted a persistent lack of change availability during peak travel hours, which stalls the boarding process and creates immense psychological friction.

As one respondent noted, *"Most of the times conductor replies with the statement 'I don't have any change I'll return it to you as soon as I get it' and it stresses me throughout my journey."* The severity of this issue is underscored by the fact that multiple non-regular commuters reported abandoning public transit entirely in favor of private ride-sharing simply to avoid the daily hassle of carrying exact cash. This highlights the necessity for a frictionless, rapid payment mechanism that removes physical currency exchange entirely.

3.3.2 Friction 2: Trust, Transparency, and Discount Verification. Respondents indicated significant behavioral challenges regarding financial trust. Users frequently cited the risk of arbitrary overcharging and inconsistent fare collection. One commuter recalled, *"Different fares each time for same route and conductors insisting*

that they're charging the right fare." Another highlighted the predatory nature of manual collection on vulnerable populations, stating, *"They took 50 rupees instead of 25 when I was new in Kathmandu city."*

Furthermore, marginalized groups and students (who comprised the majority of the eligible discount pool in our study) reported consistent anxiety in verifying their government-mandated discount eligibility. Even when presenting valid identification, commuters face pushback. A student noted, *"Sometimes the conductors are suspicious on whether the ID card is authentic or not."* These responses validate the need for an automated, location-based fare calculation system that removes human discretion and instantly applies Role-Based Access Control (RBAC) discount profiles.

3.3.3 Friction 3: The Digital Divide and Network Fragility. While users expressed a high desire for cashless convenience, they noted the limited adoption and reliability of current mobile payment systems in transit. Respondents expressed frustration over the disconnect between Nepal's retail digital economy and its transit infrastructure. One commuter explained, *"In this world of digitalization many people don't carry cash so we face issue."*

Attempting to scan a retail QR code requires continuous mobile data, and users reported high stress levels associated with transactions failing mid-commute due to Nepal's patchy urban broadband. The findings suggest that relying on the commuter's personal mobile data plan creates a digital divide, necessitating an "offline-first" system that shifts the connectivity burden to the vehicle's infrastructure.

4 Socio-Technical System Design

Traditional engineering approaches to digital payments often prioritize maximum processing power and assume continuous cloud connectivity [18]. However, deploying public transit infrastructure in the Global South requires a different paradigm. The TransitPay ecosystem was engineered using a Systems Engineering Design Methodology specifically adapted for resource-constrained environments. By harmonizing affordable IoT hardware, a Host Card Emulation (HCE) [2] enabled mobile client, and a Node.js server, the system is designed to prioritize commuter resilience and digital equity over raw computational complexity.

4.1 Design Rationale and Requirements

Rather than forcing users to adapt to a standard retail digital wallet, TransitPay is designed to adapt to the physical and economic realities of Kathmandu's transit environment. We translated the three primary socio-technical frictions identified in our formative study (Section 3) directly into three core architectural requirements:

Requirement 1: Edge-Based Resilience (Solving the Network Fragility). Because commuters reported severe stress when mobile wallets failed in network blind spots (Friction 3), the system cannot rely on the user's smartphone for real-time cloud authorization. The architecture must shift the connectivity burden to the vehicle's Point-of-Sale (POS) terminal and utilize local data queuing to process transactions entirely offline.

Requirement 2: Hardware Affordability and Speed (Solving the Cash Bottleneck). To eliminate the delays of manual cash handling and QR scanning (Friction 1), the system requires sub-second transaction speeds. To enable rapid hardware adoption by independent, low-resource microbus operators,

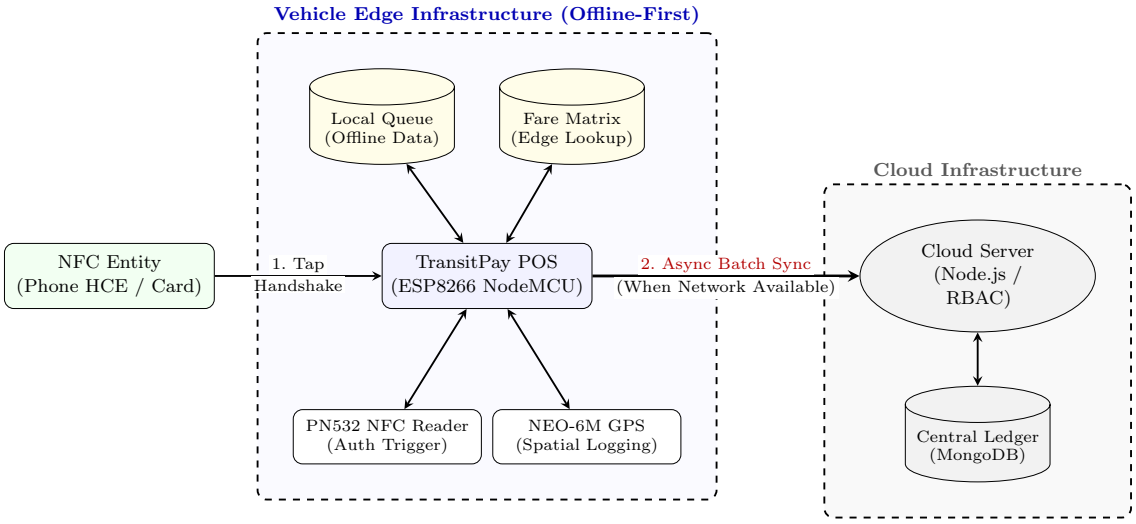


Fig. 1. TransitPay system block diagram detailing the division between offline-capable edge infrastructure and asynchronous cloud synchronization.

the POS terminal must be built from ubiquitous, low-cost IoT microcontrollers rather than expensive proprietary enterprise validators.

Requirement 3: Automated Transparency (Solving the Trust Deficit). To alleviate passenger anxiety regarding inconsistent manual cash collection and discount disputes (Friction 2), the system must remove human calculation entirely. The architecture requires integrated spatial telemetry to automate distance-based billing fairly, transparently, and objectively.

4.2 Hardware Architecture for Affordability and Speed

To meet the requirement of hardware affordability and rapid processing (Requirement 2), the Point-of-Sale (POS) terminal must bypass expensive, proprietary enterprise validators. The TransitPay POS is built around the ESP8266 (NodeMCU) microcontroller. This specific module was selected for its extreme affordability, low power profile, and integrated Wi-Fi capabilities. By assembling the POS organically from ubiquitous IoT components, the initial capital expenditure deployed per vehicle is drastically lowered, encouraging adoption among independent microbus operators.

The ESP8266 operates in tandem with a PN532 NFC reader. To ensure that lower-income commuters are not digitally excluded by a smartphone-only mandate, the system supports a dual-medium approach. It utilizes Host Card Emulation (HCE) to empower standard Android smartphones to function as transit cards, while concurrently supporting low-cost, physical NFC smart cards for users without digital devices. The architectural block diagram is shown in Figure 1.

Table 1. Machhapokhari to Ratnapark Route Fare Matrix (NPR) - Student Rate

From / To	Machhapokhari	Balaju	Nayabazar	Sohrakhutte	Thamel	Lainchaur	Jamal	Ratnapark
Machhapokhari	–	20	20	20	25	25	25	25
Balaju	20	–	20	20	20	25	25	25
Nayabazar	20	20	–	20	20	20	25	25
Sohrakhutte	20	20	20	–	20	20	20	25
Thamel	25	20	20	20	–	20	20	20
Lainchaur	25	25	20	20	20	–	20	20
Jamal	25	25	25	20	20	20	–	20
Ratnapark	25	25	25	25	20	20	20	–

4.3 Automated Transparency via the Fair Fare Model

To cultivate necessary trust among a commuter base historically reliant on physical cash, the system implements a "Fair Fare Model" (FFM). This model addresses the trust deficit and discount disputes (Requirement 3) by completely removing human calculation from the billing process. The ecosystem integrates a NEO-6M GPS module directly into the POS terminal. This module generates precise spatial telemetry, allowing transparent, distance-based fare calculations entirely at the edge.

The operational logic of the FFM relies on a three-step transaction pipeline integrated with the onboard hardware:

(1) **Tap-In:** Upon boarding, the passenger taps their NFC entity against the PN532 module. The NEO-6M simultaneously logs the geographic coordinates of the boarding stop.
(2) **Tap-Out:** When the passenger reaches their destination and exits, the system captures the alighting stop coordinates.
(3) **Lookup and Deduction:** The system matches the coordinate pair against a pre-loaded stop database on the ESP8266. The corresponding fare is retrieved from a fixed matrix and deducted instantly.

Real-World Mapping and Fare Matrix. To validate this edge-based lookup mechanism, we physically mapped several high-traffic urban transit corridors in Kathmandu, including the *Pulchowk* to *Ratnapark* and *Machhapokhari* to *Ratnapark* routes. Table 1 demonstrates the localized matrix programmed into the ESP8266 for the *Machhapokhari* route. This specific eight-stop corridor is presented as our primary example because its length and granularity better illustrate the algorithmic complexity of the distance-based billing compared to shorter routes. By storing this matrix locally, the terminal can instantly cross-reference the boarding and alighting coordinates to deduct the exact government-mandated fare without requiring any cloud processing or conductor intervention.

4.4 The Commuter Interaction Pipeline

The resulting integration of this hardware and spatial tracking creates a frictionless pipeline designed to preserve commuter dignity and eliminate boarding bottlenecks. The pipeline has three phases, also shown in Figure 2.

Phase 1: Frictionless Boarding. Upon entry, commuters intuitively wave their NFC device near the terminal. The terminal instantly authenticates the card and verifies sufficient minimum balance locally,

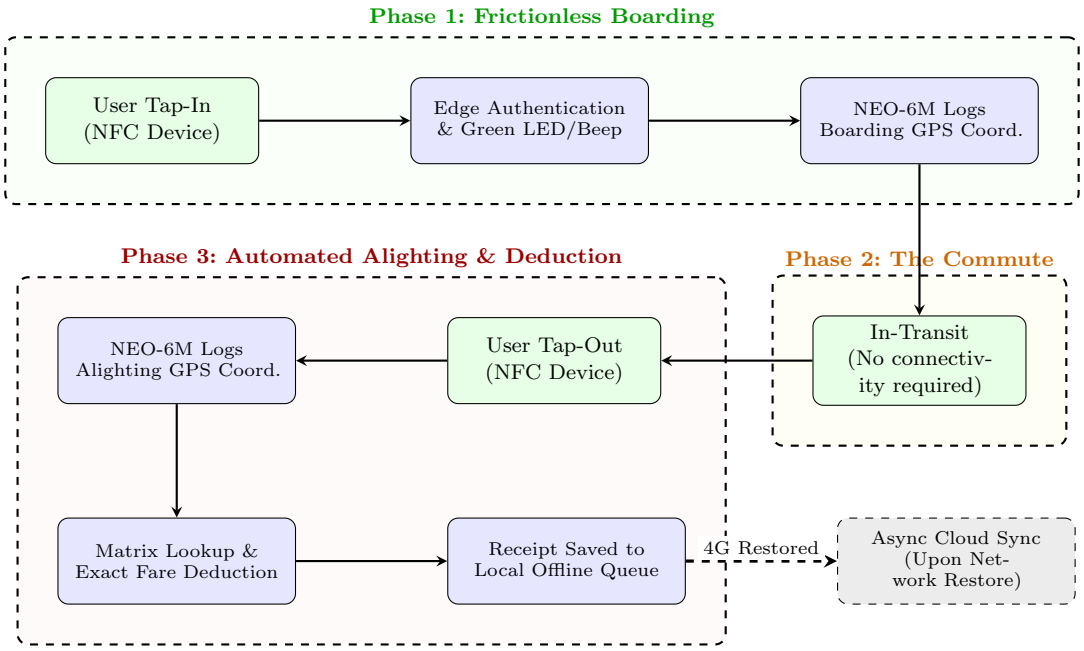


Fig. 2. The commuter interaction pipeline. The diagram contrasts intuitive user actions with simultaneous edge-computing tasks (GPS logging, offline queuing), ensuring a seamless experience regardless of cellular availability.

shielding the commuter from potential cellular network dropouts during boarding. To ease accessibility for all users, the system provides immediate audio-visual confirmation, such as a localized beep and a green LED indicator. Concurrently, the GPS module silently logs the precise geographical coordinates of the boarding event to initiate the equitable cycle.

Phase 2: The Commute. Passengers seamlessly traverse their transit route without the disruption of a conductor physically navigating a severely overcrowded aisle to collect loose cash. This stage significantly enhances passenger comfort, alleviating the persistent stress and security concerns associated with manual cash handling during peak urban commuting hours.

Phase 3: Automated Alighting and Reconciliation. As the commuter exits the microbus, a secondary brief tap finalizes the trip footprint. The exact required amount is deducted, averting overcharging and verbal disputes. If a commuter fails to tap out upon exiting, the system defaults to deducting the maximum route fare to incentivize compliance. Crucially, the finalized receipt is queued locally. When the transit vehicle transitions from a cellular dead zone back into a strong network range, the offline transaction logs cached resiliently in the ESP8266 are seamlessly synchronized with the cloud ledger to guarantee an airtight digital audit trail.

5 System Evaluation and Results

We designed a mixed-environment evaluation framework to assess whether TransitPay successfully resolves the socio-technical frictions identified in our formative study. The prototype was evaluated against both controlled laboratory benchmarks and real-world urban transit conditions.

5.1 Experimental Setup and Baseline

The TransitPay ecosystem was deployed entirely within a sandboxed testing framework. Initial hardware and cryptographic handshakes were stress-tested in a controlled laboratory setting to establish baseline performance metrics. Following the lab tests, the ESP8266 Point-of-Sale terminals were deployed for field evaluation at the primary transit hubs of specifically mapped real-world public transit corridors in Kathmandu.

Because local transit syndicate regulations restricted the installation of prototype hardware inside operational, passenger-carrying microbuses, the field tests were conducted curbside at the exact boarding and alighting coordinates. We evaluated the system across two distinct environments: the *Machhapokhari* to *Ratnapark* corridor and the high-traffic *Pulchowk* to *Ratnapark* corridor. By bringing the sandboxed hardware to these specific geographic hubs, we successfully validated the Fair Fare Model's spatial telemetry and local matrix lookup against actual GPS coordinates. Furthermore, this curbside testing allowed us to measure transaction latency and offline queuing capabilities using the live, often congested 4G LTE cellular conditions commuters experience exactly at the point of boarding.

To establish a comparative baseline for our results, existing digital wallets of Nepal (specifically *eSewa* and *Khalti*) were simultaneously evaluated in their live production modes. Because these applications rely entirely on a QR-based payment architecture designed for stable retail environments, their performance during transit was assessed to quantify the exact gap between the current digital payment capabilities available to Kathmandu commuters and the reality of the transit environment.

5.2 Evaluation Metrics

The comparative evaluation between the sandboxed TransitPay architecture and the production-level QR wallets was measured across five core metrics, each directly tied to the human frictions outlined in Section 3:

- **Latency (Addressing the Boarding Bottleneck):** Measured as the exact time frame between the initial user physical interaction (tap or scan) and the output display rendering a success or error message from the server.
- **Reliability (Addressing Network Fragility):** Defined as the system's ability to consistently process transactions without failure, resulting in an overall success rate percentage under varying bandwidth conditions.
- **Fault Tolerance:** The capacity of the edge-based hardware to queue and handle internal or external failures, such as sudden cellular network drops or server downtime, without interrupting the user experience.
- **Security:** An assessment of the cryptographic vulnerability of the data payload during the transmission between the sender (the commuter's device) and the receiver (the POS terminal).

Table 2. Payment System Performance Comparison

Factor	TransitPay	Esewa/Khalti QR
Method	NFC Tap-in and Tap-out	Receiver's QR scanned by user
Reliability	83.33% success rate on 6 Mbps	100% success rate on 8 Mbps
Latency	3 seconds average response time	35 seconds average response time
Fault Tolerance	Local queuing for connectivity issues	Direct failure (no offline service)
User Connectivity	Not required	Essential to use services
POS Security	Encrypted communication only	Static QR code (data exposed)

- **Usability and Automated Accuracy (Addressing the Trust Deficit):** A qualitative and quantitative measure of how conveniently the system processes the fare without human intervention, ensuring the correct financial amount is deducted based on the spatial route data.

5.3 Performance Results and Interpretation

The comparative performance between TransitPay and the baseline QR-based digital wallets (*eSewa* and *Khalti*) demonstrates significant operational advantages in resource-constrained transit environments.

5.3.1 Latency and the Boarding Bottleneck. In informal transit environments, cumulative boarding delays directly exacerbate physical crowding. Transaction speeds were measured across varying typical network conditions to quantify TransitPay's impact on mitigating these delays:

- **Moderate Connectivity (6 to 8 Mbps):** The system processed tap-and-go transactions in an average of 2.5 to 3.3 seconds.
- **Constrained Connectivity (2 to 5 Mbps):** Operating resiliently on edge-queued data within cellular dead zones, transaction times increased to roughly 6 to 7 seconds. This crucially averted total system failure and avoided a fallback to cash modes.

Compared to the 35 seconds often required to properly orient and scan optical QR codes within moving, tightly packed vehicles, TransitPay drastically diminishes entry bottlenecks.

5.3.2 Reliability and Fault Tolerance. While traditional digital wallets struggle with patchy broadband, TransitPay provides continuous service availability. As detailed in Table 2, QR systems experience direct failure during network drops. Conversely, TransitPay achieves complete fault tolerance through its local queuing mechanism on the ESP8266. When a network failure is detected, the device locally stores the encrypted transaction intent (ID, timestamp, and GPS coordinates). Upon re-entering a high-bandwidth zone, the system automatically initiates an asynchronous synchronization process with the cloud server.

5.3.3 Security Architecture. The security of the TransitPay ecosystem is enforced through a multi-layered cryptographic approach. The POS terminal issues an Application Protocol Data Unit (APDU) challenge to the user's NFC entity. The entity signs the payload using its unique private key, which is decrypted by the POS using a corresponding embedded public key. Furthermore, the physical constraints of NFC technology limit communication to approximately 4 cm, mitigating remote eavesdropping and man-in-the-middle attacks.

Table 3. Tap-in and Tap-out Edge Lookup Test

Tap-In Hub	Tap-Out Hub	Outcome
Thamel	Machhapokhari	Success (25 rupees deducted)
Ratnapark	Thamel	Success (20 rupees deducted)
Ratnapark	Pulchowk	Failure (Route mismatch)
Pulchowk	Ratnapark	Success (25 rupees deducted)
Machhapokhari	Balaju	Failure (GPS coordinate error)
Machhapokhari	Ratnapark	Success (25 rupees deducted)

5.3.4 Fair Fare Model Accuracy. By seamlessly automating fare calculations via the GPS-linked Fair Fare Model, TransitPay enforces transparent pricing frameworks. During field evaluations, the automated schema successfully bridged spatial mapping to equitable financial deductions. Table 3 illustrates the system's ability to accurately cross-reference coordinates or reject invalid routes.

5.3.5 Usability and Mitigating the Trust Deficit. The usability of TransitPay was evaluated qualitatively through its capacity to resolve the psychological and physical burdens identified in our formative study. In the current cash-based or QR-based environment, passengers must interact extensively with conductors or navigate complex mobile interfaces while balancing in a moving vehicle. TransitPay simplifies this process to a single, intuitive physical gesture.

Feedback from the real-world deployment indicates that the minimal interaction time (under 3 seconds) and the complete elimination of the smartphone internet requirement are the primary drivers of user adoption. By automating fare calculations, the system effectively removes the stress of verbal fare disputes (Friction 2) and the inconvenience of fumbling for exact physical change (Friction 1). This frictionless tap-and-go experience preserves passenger dignity, particularly for marginalized groups who rely on automated discount verification, while simultaneously accelerating the overall throughput of the transit network.

5.4 Economic Viability and Hardware Affordability

A persistent barrier to digitizing informal transit in the Global South is the prohibitive capital expenditure required for proprietary enterprise transit validators [40]. To ensure that TransitPay is economically viable for independent microbus operators in Kathmandu, the system was intentionally constrained to utilize low-cost, off-the-shelf IoT components.

Table 4 details the unit production cost of a single TransitPay Point-of-Sale edge terminal. The total hardware expenditure is approximately 3,000 NPR (roughly $22 USD).

By reducing the capital requirement by over 90 percent compared to traditional transit hardware, this architecture significantly lowers the barrier to entry for widespread fleet adoption. Furthermore, because the architecture relies on the users' existing smartphones via Host Card Emulation (HCE), the system avoids the massive logistical costs associated with minting and distributing millions of proprietary closed-loop smart cards.

Table 4. Unit Production Cost of the Edge POS Terminal

Component	Functionality	Cost (NPR)
ESP8266	Core microcontroller with Wi-Fi	800
PN532	NFC read/write module	600
NEO-6M	GPS spatial telemetry	500
LCD Screen	Audio-visual user feedback	800
Housing	3D printed protective case	300
Total	**Deployed edge terminal**	**3000**

6 Discussion and Implications

6.1 Alignment Between Identified Problems and Proposed Solutions

The TransitPay ecosystem was engineered as a direct technological response to the specific pain points identified in our user survey (Section 3). Beyond solving immediate technical frictions, this design demonstrates how offline-first, edge-based computing can operationalize principles of digital equity and sustainable urban mobility in resource-constrained contexts. The alignment between the reported real-world challenges and the proposed technical features is summarized through the following mappings:

Cash Dependency and Change Unavailability: The survey highlighted significant frustration regarding the lack of change during peak travel hours, a daily indignity that compounds time poverty for low-income commuters. TransitPay eliminates the need for physical currency by utilizing NFC-enabled smartphones or tap-to-pay smart cards. Beyond mere convenience, this exact precision in fare settlement removes the burden of cash management from both passengers and conductors, actively reducing the daily psychological stress and interpersonal friction between riders and transport staff while enabling smoother, more sustainable transit flows.

Fare Inconsistency and Overcharging: Users reported a lack of transparency in fare calculation, which disproportionately affects vulnerable populations such as new city residents and migrant workers. The Fair Fare Model, integrated with NEO-6M GPS modules, automates the billing process by calculating fares between exact Tap-In and Tap-Out hubs. By enforcing the government-mandated fare matrix (Table 1), the system eliminates human error and predatory pricing, thereby restoring fundamental financial trust in the informal transit ecosystem and modeling how automated transparency can serve as a form of structural justice in digital public infrastructure.

Discount Eligibility Verification: One of the major hurdles identified was the difficulty and embarrassment of verifying student or elderly discounts in real-time. The TransitPay system handles this at the Check-in Authentication phase (Phase 1), where server-side role-based access control (RBAC) automatically applies the appropriate discount profile without human intervention. This invisible verification preserves passenger dignity by removing the need for physical ID confrontation in crowded spaces, demonstrating how computational governance can operationalize social equity at scale.

Crowding and Manual Latency: The identified discomfort of manual transactions in crowded buses is addressed by the 91.4% reduction in transaction latency. By replacing manual QR scanning with a sub-3-second NFC tap, the system prevents passenger stacking at entry points. On a macro level, reducing

these cumulative boarding bottlenecks combats the time poverty experienced by essential workers who rely on these transit arteries, indirectly supporting more sustainable urban mobility patterns by making public transit more competitive with private alternatives.

Network Resilience: While traditional digital wallets struggle with patchy broadband, TransitPay's local queuing mechanism allows complete fault tolerance (Table 2) via offline transaction capture. This design choice actively promotes digital equity by ensuring that commuters without consistent or expensive mobile data plans are not excluded from the modernization of public infrastructure.

This mapping shows that our proposed solution effectively bridges both the structural infrastructure gaps and the behavioral trust deficits in Nepal's current public transit landscape. More broadly, it offers a concrete example of how socio-technical design, grounded in local frictions but guided by principles of equity and resilience, can transform informal transit systems into platforms for sustainable, inclusive urban computing.

6.2 Persisting Infrastructural Barriers

While the TransitPay prototype demonstrates strong technical viability, scaling such an intervention across informal transit networks presents challenges that extend well beyond engineering constraints. These barriers are rooted in the deeply entrenched political economy, financial practices, and material realities of Kathmandu's transport ecosystem.

The Political Economy of Route Syndicates and Revenue Opacity: A primary constraint lies in the governance structure of Kathmandu's informal transit system. Public transport is heavily cartelized by decentralized route syndicates and independent vehicle owners [30]. These actors collectively regulate route access and operations through an informal, cash-based micro-economy [39]. Within this structure, vehicle owners typically impose a fixed daily revenue target on drivers and conductors. Any earnings beyond this threshold become the crew's take-home income. This arrangement directly incentivizes overcrowding and aggressive passenger acquisition practices, as maximizing onboard passengers is essential for workers to secure a livable wage.

TransitPay fundamentally disrupts this equilibrium. By generating an immutable digital ledger of passenger boarding and alighting events, the system introduces full financial transparency into a previously opaque ecosystem [36]. This shift transfers visibility and control over revenue flows to vehicle owners, effectively eliminating the informal margins that transport workers rely upon. While such transparency enhances accountability, it simultaneously risks economically disenfranchising the very actors who sustain daily operations. This tension reflects a broader challenge in computing for sustainable societies: digitization can enable systemic efficiency while redistributing economic power in ways that disadvantage vulnerable labor groups. Scaling TransitPay will require participatory design approaches that actively negotiate these trade-offs and safeguard worker livelihoods.

Cash Flow Settlement Latency: Beyond governance and labor dynamics, financial liquidity introduces an additional barrier to adoption. Informal transit operations depend on immediate access to daily cash flows to cover essential expenses, including fuel, syndicate fees, and routine maintenance. Transitioning to a digital payment system introduces unavoidable settlement delays. Although TransitPay records transactions instantly at the edge and synchronizes data when connectivity is available, the actual transfer of funds to operators' accounts may take 24 to 48 hours [24].

For operators functioning on narrow margins, this delay creates a critical cash flow bottleneck. The absence of same-day liquidity can disrupt operations and discourage system adoption. Addressing this constraint requires complementary financial mechanisms, such as guaranteed instant settlements or bridge financing systems, that align digital payment infrastructures with the real-time economic needs of informal operators.

Automotive Power Instability and Hardware Degradation: In addition to socio-economic constraints, physical deployment introduces significant technical challenges. Integrating IoT hardware into aging vehicle fleets exposes TransitPay to unstable electrical environments and harsh operating conditions. The ESP8266-based terminals require a consistent power supply, yet many microbuses rely on degraded 12-volt systems prone to voltage fluctuations and component failures [25, 31].

Environmental factors further compound these issues. Vehicles operating across Kathmandu's routes are subjected to continuous vibration, high dust levels during dry seasons, and elevated humidity during monsoons [13]. While the use of low-cost, off-the-shelf components reduces upfront costs and supports local repair ecosystems, their long-term durability under such conditions remains uncertain [17]. These challenges underscore a key principle in infrastructure-aware system design: technologies must be engineered not for ideal conditions, but for the constraints of the environments in which they are deployed.

Third-Party API Dependency: Finally, the system's reliance on external institutional infrastructure introduces additional uncertainty. TransitPay depends on third-party APIs, such as those associated with national digital identity frameworks, to enable automated verification for discounted fares. However, access to such infrastructure is not guaranteed. Downtime, network latency, or bureaucratic barriers to integration can disrupt the Check-in Authentication process. If the edge system is unable to periodically update its local verification data or cryptographic credentials, the reliability of automated validation deteriorates. In such cases, the system risks reverting to manual verification processes, thereby undermining one of its core value propositions of efficient and equitable access for marginalized commuter groups.

6.3 Strategic and Technical Recommendations

To enable sustainable deployment and scalability of offline-first transit payment systems in the Global South, we propose a multi-faceted approach spanning technical, operational, and policy interventions:

(1) **Ruggedized Hardware for Local Repairability:** Future POS terminals should incorporate ruggedized enclosures and integrated power regulation to withstand automotive instability and environmental stress. Localized battery backups or supercapacitors can protect ESP8266 microcontrollers from voltage fluctuations, while modular designs support maintenance by local technicians rather than vendor dependency.

(2) **Phased Rollout with Operator Incentives:** Deployment should follow a phased model to reduce resistance from transit syndicates and address cash flow constraints. Early stages must prioritize rapid settlement cycles, such as daily batch transfers. In parallel, performance-based micro-incentives for conductors can encourage adoption by aligning financial benefits with increased usage of cashless transactions.

(3) **Universal NFC Integration in Public IDs:** National identification systems should standardize the inclusion of NFC chips in student and senior citizen ID cards. This removes dependence on personal smartphones and enables inclusive access to automated discount verification.

(4) **Open API Ecosystems for Transit Data:** Transport authorities should publish and maintain standardized, geographically mapped fare matrices through open APIs. This shared infrastructure enables decentralized edge systems to perform accurate distance-based billing without requiring frequent manual updates or proprietary integrations.

7 Limitations

This study has a few technical as well as non-technical limitations. On the technical side, Apple doesn't support the development of Host Card Emulation (HCE) on the iOS platform outside of the European Economic Area (EEA). Furthermore, this means that developers in regions such as Nepal must rely on Apple Pay or Apple Wallet for contactless NFC transactions, rather than implementing their own HCE-based payment or ticketing solutions directly on iOS. [16] However, this makes more difficult to access the Host Card Emulation in IoS devices as the Apple Pay payment feature itself is disabled for Nepal-registered Apple IDs and Nepali cards. Additionally, the system lacks on-ground testing in real-world public transport environments; specifically, the prototype could not be tested with the surveyed users who were facing the identified problems, which could have provided valuable insights into its performance and usability under actual operating conditions. From a non-technical perspective, the research was constrained by budgetary resources, which prevented a more longitudinal analysis of the observed phenomena.

8 Conclusion

The modernization of public transit in the Global South cannot depend on simply replicating retail digital wallet systems. Our formative study and field evaluations in Kathmandu reveal that imposing cloud-dependent payment infrastructures on resource-constrained transit networks worsens digital exclusion and increases passenger friction. TransitPay offers a human-centered alternative for urban mobility. By shifting computational and connectivity demands from commuters' smartphones to affordable edge infrastructure, the system reconciles digital convenience with on-the-ground realities. The integration of ubiquitous IoT hardware with a localized Fair Fare Model shows that high latency and arbitrary cash disputes are not inevitable features of informal transit. Achieving a 91.4 percent reduction in boarding transaction times and ensuring full fault tolerance in cellular dead zones, the architecture protects both the economic margins of operators and the daily dignity of marginalized commuters. Hence, TransitPay demonstrates that true digital equity in sustainable societies emerges not by forcing commuters to conform to fragile technologies, but by creating resilient systems that respond to the realities of everyday life.

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